DS 7400 DEEP LEARNING / HW3 EXPERIMENT SUMMARY

Can a second decision reduce mesh classification errors?

Prediction before training. Mesh Atlas classifies a synthetic mesh as an ellipsoid, cuboid, cylinder or torus. The HW2 reference model misclassified only five of 600 test objects, but four errors crossed the cuboid–cylinder boundary. I expected that asking the classifier to distinguish these shapes explicitly during training would reduce errors in both directions combined. A risk is that the 24 descriptors discard the relevant surface detail. In that case, emphasizing the distinction could amplify noise.

Baseline and design. The balanced dataset contains 1,600 training, 400 validation and 600 test objects, with variation in pose, scale, proportions and vertex noise. The architecture is identical at inference in all conditions.

$$\underbrace{\text{24 standardized descriptors}}_{\text{train-set mean and scale}} \longrightarrow \text{48 ReLU} \longrightarrow \text{48 ReLU} \longrightarrow \text{4 logits}$$

For a batch of size B, the baseline minimizes mean four-way cross-entropy L_4. The modified objective reuses two existing logits. If P is the set of cuboid and cylinder examples in the batch, its loss is

$$L_4 + \frac{1}{B}\sum_{i \in P} -\log \frac{e^{z_{i,y_i}}}{e^{z_{i,\text{cuboid}}} + e^{z_{i,\text{cylinder}}}}.$$

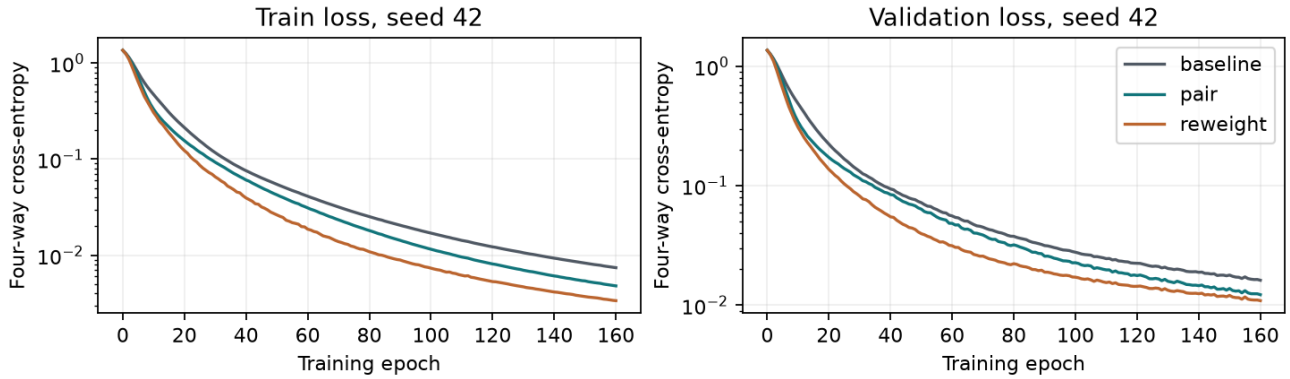
This extra term pushes the correct logit above its specific rival even when the other two classes take probability mass. The control uses the same examples and coefficient but replaces the two-way denominator with the four-way denominator. This doubles the ordinary loss contribution of each targeted object and tests whether extra emphasis alone explains an improvement.

Controlled comparison. Each condition used SGD for 160 epochs, batch size 64, learning rate 0.03 and weight decay 0.0001. Within each seed, all three runs started from identical weights and saw the same batch order. We selected checkpoints by ordinary four-way validation loss. The primary outcome counts cuboids predicted as cylinders plus cylinders predicted as cuboids among the 300 test objects of those classes.

Condition	Pair errors / all errors by seed 42	43	44	Mean test accuracy
HW2 baseline	4 / 5	3 / 3	4 / 4	99.33%
Two-way auxiliary loss	2 / 3	2 / 2	3 / 3	99.56%
Four-way reweighting	3 / 4	2 / 2	3 / 3	99.50%

OBSERVATIONS / ERROR ANALYSIS

Extra practice helped. Its form mattered less.



Where the gains occurred. At seed 42 the baseline mislabeled one cuboid as a cylinder and three cylinders as cuboids. The two-way loss eliminated the first error and reduced the second count to two. One cylinder was still called a torus. Across seeds, the two-way objective reduced the targeted error count every time, with no increase in errors outside the pair. Yet it beat simple reweighting by only one object at seed 42 and tied it at seeds 43 and 44.

Optimization and trade-offs. The seed-42 baseline reproduced HW2's 99.17% test accuracy. All three models learned smoothly overall. At that seed the two-way objective reduced ordinary training/validation loss from the baseline's 0.0074/0.0162 to 0.0048/0.0122. The control achieved 0.0034/0.0109 but made one more test error than the two-way model. A lower average cross-entropy measures probability assigned to the true labels, while an error count changes only when the highest-scoring class changes. Validation losses remained above training losses, but neither the curves nor the selected epochs (157–160 across runs) suggest severe early overfitting. All models retain 3,748 parameters. Only training adds work to evaluate the extra loss.

What the comparison establishes. Extra practice on these two categories accounts for most of the observed reduction. The evidence for a specific benefit from isolating the competing logit is weaker. The same 600 test meshes were reused across seeds, and HW2's test errors motivated the choice of pair. These are exploratory comparisons on familiar objects. A next experiment would freeze both objectives and evaluate them on freshly generated meshes, stratifying errors by aspect ratio to test whether the remaining failures cluster at particular proportions.